I. Definitions and glossary

I.1 Introduction

Differential equations can be used to give a mathematical description of the evolution of phenomena studied in many fields such as physics, biology, economics, etc.
For example : the determination of the speed of a falling body, the discharge of a capacitor in an electric circuit, population dynamics, etc.

We call an **Ordinary Differential Equation** of order n or (ODE) an equation of the form :

$$F(x, y(x), y'(x), y''(x), \dots, y^{(n)}(x)) = 0, \quad \forall x \in I,$$

where I is an interval of $\mathbb{R}$, F is a function of several variables and y is an unknown function defined from I into $\mathbb{R}$.
It is about an equation that relates an unknown function y of a real variable x with its successive derivatives.
An ODE is of order n if it contains derivatives of y up to order n.
It is usual to denote this unknown function by y instead of $y(x)$ in the differential equation.
For example :

1. The equation $x^2y' + (x-1)y^2 - 4x = 0$ is a first-order ODE.
2. The equation $y'' + xy' = \ln x$ is a second-order ODE.
3. The equation $-y^{(3)} + 2x(y-1) = e^x$ is an ODE of order 3.

The first part of this chapter deals with **linear differential equations of the first order** and the second with **linear differential equations of the second order**.

II. First Order Linear Differential Equation

II.1 Definition of first-order linear differential equation

Definition 1.
A first-order linear equation is an equation given by the form :

$$\boxed{a(x)y' + b(x)y = f(x) \qquad (E)},$$

where :

- *a and b are given continuous functions on an interval I of $\mathbb{R}$ with $a(x) \neq 0$, for all $x \in I$ and they are called coefficients of equation (E).*
- *f is a function given on I called second member of equation (E).*
- *y is an unknown function of class $\mathcal{C}^1$ on I to be determined.*

A function f is said to be of class $\mathcal{C}^1$ on some interval I of $\mathbb{R}$, if f is differentiable on I and its derivative f' is continuous on I.
The notion of successive derivatives and class $\mathcal{C}^n$ ($n \in \mathbb{N}$), will be presented in detail in the next chapter.

Definition 2.
The first-order linear differential equation ***homogeneous*** *or* ***equation without second member associated with*** (E)*, the equation of the form :*

$$\boxed{a(x)y' + b(x)y = 0 \qquad (H)}$$

Example 1.

1. $(x^2 - 1)y' + xy = e^x$. *This is a first-order linear ODE with second member.*
2. $(chx)y' + (shx)y = 0$. *This is a first-order linear ODE without second member.*

II.2 Method for solving a first-order linear differential equation

II.2.1 Solving the homogeneous equation

Let (H) be the following equation without second member or homogeneous:

$$a(x)y' + b(x)y = 0 \qquad (H)$$

The following theorem gives all the solutions of the homogeneous equation (H).

Theorem 1.
Let a and b be two continuous functions on an interval I with $a(x) \neq 0$, for all $x \in I$.
The solutions on I of the homogeneous differential equation (H) are given by :

$$\boxed{y_h(x) = \lambda e^{-F(x)},}$$

where $F(x)$ is a primitive of $\dfrac{b(x)}{a(x)}$ and λ is a real constant.

Example 2.

1. *Consider the homogeneous equation : $xy' - 2y = 0$. For $x \neq 0$, according to Theorem 2, the solutions of this equation on $\mathbb{R}^\star$ are given by :*

$$\boxed{y_h(x) = \lambda e^{\int \frac{2}{x}\,dx} = \lambda x^2, \quad \lambda \in \mathbb{R}.}$$

2. *Consider the homogeneous equation : $(\cos x)y' + (\sin x)y = 0$.*
For $\cos x \neq 0$, i.e. for $x \in \mathbb{R} - \left\{ \pm \frac{\pi}{2} + 2k\pi,\ k \in \mathbb{Z} \right\}$, according to Theorem 2, the solutions of this equation on $\mathbb{R} - \left\{ \pm \frac{\pi}{2} + 2k\pi,\ k \in \mathbb{Z} \right\}$ are given by :

$$\boxed{y_h(x) = \lambda e^{\int -\frac{\sin x}{\cos x}\,dx} = \lambda e^{\ln(|\cos x|)} = \lambda |\cos x|, \quad \lambda \in \mathbb{R}.}$$

II.2.2 General solution of a first-order linear differential equation

For first-order linear differential equations with a second member, the following theorem gives the form of the general solution.

Theorem 2.
Let a, b and f be three continuous functions on the interval I with $a(x) \neq 0$, for all $x \in I$. The solutions of differential equation (E) are functions of the form :

$$\boxed{y_g(x) = y_h(x) + y_p(x)}$$

où

- y_h *represents the solutions of the homogeneous equation associated with the equation* (E): $a(x)y' + b(x)y = 0$.
- y_p *is* ***the particular solution*** *of the equation* (E).
- y_g *is called* ***general solution*** *of the equation* (E).

Remark 1.
Through this theorem, we notice that to solve the differential equation (E), we go through two steps.

- ▷ *Solving the the homogeneous equation (H) associated with the equation (E).*
- ▷ *The determination of the particular solution of the equation (E).*

II.2.3 Particular solution

Definition 3.
We call ***particular solution*** *of differential equation (E), any function of class C^1 on I and satisfying equation (E).*

Example 3.

- *The function $y_p(x) = x^3$ is a particular solution of the the differential equation:*

$$xy' - 2y = x^3. \qquad (1).$$

So, using Theorem 1 and the first homogeneous equation of example 2, we deduce that the general solution of (1) is given by :

$$\boxed{y_g(x) = y_h(x) + y_p(x) = \underbrace{\lambda x^2}_{y_h(x)} + \underbrace{x^3}_{y_p(x)}, \quad \forall x \in \mathbb{R}^\star, \quad \lambda \in \mathbb{R}.}$$

- *The function $y_p(x) = \sin x$ is a particular solution of the differential equation:*

$$(\cos x)y' + (\sin x)y = 1. \qquad (2)$$

So, going back to to Theorem 1 and the second homogeneous equation of example 2, we deduce that the general solution of (2) is given by :

$$\boxed{y_g(x) = y_h(x) + y_p(x) = \underbrace{\lambda|\cos x|}_{y_h(x)} + \underbrace{\sin x}_{y_p(x)}, \quad \forall x \in \mathbb{R} - \left\{ \pm \frac{\pi}{2} + 2k\pi,\ k \in \mathbb{Z} \right\}, \quad \lambda \in \mathbb{R}.}$$

In the general case where a particular solution is not simple to deduce it, we use **the variation of the constant method**. This is a method for determining the particular solution of the differential equation with second member (E) (or the complete equation), knowing the general solution of the homogeneous equation (without second member) y_h. (See course supplement 1)

II.2.4 The variation of the constant method:

Let the differential equation

$$a(x)y' + b(x)y = f(x), \qquad (E)$$

where a, b and f are continuous functions on the interval I. Let $y_h(x) = \lambda e^{-F(x)}$ be the solution of the homogeneous equation $a(x)y' + b(x)y = 0$ associated with (E). Then $y_p(x) = \lambda(x)e^{-F(x)}$ is a particular solution of the equation with second member $a(x)y' + b(x)y = f(x)$, where $\lambda(x)$ is a function of class C^1 on I, satisfying :

$$\lambda^{'}(x) = \frac{f(x)}{a(x)e^{-F(x)}}.$$

Indeed : If the function $y_p(x) = \lambda(x)e^{-F(x)}$ is a solution of the complete equation $a(x)y' + b(x)y = f(x) \quad (E)$, then it verifies this equation. We have :

$y_p'(x) = \lambda'(x)\, e^{-F(x)} - \lambda(x)\, F'(x)\, e^{-F(x)} = \lambda'(x)\, e^{-F(x)} - \lambda(x)\, \dfrac{b(x)}{a(x)} e^{-F(x)} = e^{-F(x)} \Big(\lambda'(x) - \lambda(x)\, \dfrac{b(x)}{a(x)}\Big).$

Injecting the equation (E), we obtain :

$$\begin{aligned} a(x)e^{-F(x)}\lambda'(x) \cancel{-b(x)\lambda(x)e^{-F(x)}} + \cancel{b(x)\lambda(x)e^{-F(x)}} &= f(x) \\ \Longrightarrow \lambda'(x) &= \frac{f(x)}{a(x)e^{-F(x)}}. \end{aligned}$$

Thus, to determine a particular solution y_p, we calculate the primitive function of λ.

Example 4.
Let be the following differential equation:

$$y' + 2xy = 2xe^{-x^2}. \qquad (3)$$

For all x in $\mathbb{R}$, we begin by solving the homogeneous equation associated with the differential equation (3). According to Theorem 2, we have :

$$y_h(x) = \lambda e^{-\int 2xdx} = \lambda e^{-x^2}, \quad \forall x \in \mathbb{R}, \quad \lambda \in \mathbb{R}.$$

Now let's look for a particular solution y_p of equation (3).
To do this, let's use the constant variation method and set : $y_p(x) = \lambda(x)e^{-F(x)} = \lambda(x)e^{-x^2}$, so $y_p'(x) = \lambda'(x)e^{-x^2} - 2x\lambda(x)e^{-x^2}$.
By injecting $y_p(x)$ and $y_p'(x)$ into equation (3), we obtain :

$$\begin{aligned} \Big(\lambda(x)e^{-x^2}\Big)' + 2x\lambda(x)e^{-x^2} &= 2xe^{-x^2} \\ \lambda^{'}(x)e^{-x^2} \cancel{-2x\lambda(x)e^{-x^2}} + \cancel{2x\lambda(x)e^{-x^2}} &= 2xe^{-x^2}. \end{aligned}$$

Which gives :

$$\lambda^{'}(x) = 2x \Rightarrow \lambda(x) = x^2.$$

Thus, the particular solution of the equation (3) is $y_p(x) = x^2e^{-x^2}$. We conclude that the general solution on $\mathbb{R}$ of the equation (3) is given by :

$$\boxed{y_g(x) = y_h(x) + y_p(x) = \lambda e^{-x^2} + x^2e^{-x^2} = \left(\lambda + x^2\right) e^{-x^2}, \quad \lambda \in \mathbb{R}.}$$

III. Second Order Linear Differential equations

III.1 Definition of a second-order linear differential equation with constant coefficients

Definition 4.
The second-order linear differential equation with constant coefficients, is any equation of the form :

$$\boxed{a\,y'' + b\,y' + c\,y = f(x) \qquad (E)}$$

where:

- *a, b and c are real constants given by $a \neq 0$, they are called* ***coefficients of the equation*** *(E).*
- *f is a given continuous function on an interval I of $\mathbb{R}$ called* ***second member of the equation*** *(E).*
- *y is* ***an unknown function*** *of class C^2 on I, to be determined.*

A function f is said to be of class $\mathcal{C}^2$ on some interval I of $\mathbb{R}$, if f is differentiable on I and its derivative f' is also differentiable on I and its derivative f'' is continuous on I.

Definition 5.
The second-order ***homogeneous*** *linear differential equation or equation* ***without second member*** *associated with (E), the equation of the form :*

$$\boxed{a\,y'' + b\,y' + c\,y = 0 \qquad (H)}$$

IV. Method for solving a second-order linear differential equation with constant coefficients

IV.1 Solving the homogeneous equation (or equation without second member)

We're interested in solving the equation without a second member associated with the following equation (E) :

$$\boxed{a\,y'' + b\,y' + c\,y = 0 \qquad (H)}$$

The following theorem gives the solutions to the equation (H) .

Theorem 3.
The solutions of the equation without a second member (H) on an interval I are given by :

$$\boxed{y_h(x) = C_1 y_1(x) + C_2 y_2(x),}$$

where y_1 and y_2 have two non-proportional solutions (i.e. : $\dfrac{y_1}{y_2} \neq k$, for any real constant k) of the equation (H) on the interval I and C_1 and C_2 are non-proportional solutions. the equation (H) on the interval I and C_1 and C_2 are any real constants.

To determine y_h, we look for solutions of the form $y(x) = e^{rx}$ where $r \in \mathbb{R}$ or $\mathbb{C}$ to be determined. By deriving y twice, we have : $y'(x) = r\,e^{rx}$ and $y''(x) = r^2\,e^{rx}$. Then, by injecting $y(x)$, $y'(x)$ and $y''(x)$ into the equation (H), we find :

$$\begin{aligned} & a\,y'' + b\,y' + c\,y = 0 \\ \iff & \Big(ar^2 + br + c\Big)e^{rx} = 0 \\ \iff & ar^2 + br + c = 0 \qquad \text{(CE)} \end{aligned}$$

Definition 6.
Equation $ar^2 + br + c = 0$ *is called* ***characteristic equation*** *associated with the equation* (H).

This brings us back to solving the characteristic equation which is a second-degree equation whose discriminant is given by : $\Delta = b^2 - 4ac$.
Table 1 summarizes the three cases of the sign of Δ and the solutions of equation (H).

Discriminant Δ	**Roots causes of the equation (CE)**	**Solutions of equation (H)**
$\Delta > 0$	Two real roots $\begin{cases} r_1 = \dfrac{-b-\sqrt{\Delta}}{2a} \\ r_2 = \dfrac{-b+\sqrt{\Delta}}{2a} \end{cases}$	$y_h(x) = C_1 e^{r_1 x} + C_2 e^{r_2 x}, C_1, C_2 \in \mathbb{R}$.
$\Delta = 0$	A double root, $r_0 = \dfrac{-b}{2a}$	$y_h(x) = (C_1 x + C_2) e^{r_0 x}, C_1, C_2 \in \mathbb{R}$.
$\Delta < 0$	Two complex rootss $\begin{cases} r_1 = \alpha + i\beta \\ r_2 = \alpha - i\beta \end{cases}$	$y_h(x) = [C_1 \cos \beta x + C_2 \sin \beta x] e^{\alpha x}, C_1, C_2 \in \mathbb{R}$, with $\alpha = \dfrac{-b}{2a}$ and $\beta = \dfrac{\sqrt{-\Delta}}{2a}$.

Table 1: Solutions to the equation without a second member

Example 5.

1. *Let the equation without a second member be* $y'' - y = 0$.
   The associated characteristic equation (CE) *is* $r^2 - 1 = 0$ *and has two real roots* $r_1 = -1$ *and* $r_2 = 1$.
   Thus, using the table 1, *we find :*

$$\boxed{y_h(x) = C_1 e^{-x} + C_2 e^{x}, \quad C_1,\ C_2 \in \mathbb{R}.}$$

2. *Let be the equation without a second member:* $y'' + y' + y = 0$.
   The associated characteristic equation (EC) *is* $r^2 + r + 1 = 0$. *Its discriminant* $\Delta = -3$. *Then,* (CE) *has two complex roots* $z_1 = -\frac{1}{2} + i\frac{\sqrt{3}}{2}$ *and* $z_1 = -\frac{1}{2} - i\frac{\sqrt{3}}{2}$.
   Thus, using the table 1, *we obtain :*

$$\boxed{y_h(x) = e^{-\frac{1}{2}x}\Big(C_1 \cos(\frac{\sqrt{3}}{2}x) + C_2 \sin(\frac{\sqrt{3}}{2}x)\Big) \quad C_1,\ C_2 \in \mathbb{R}.}$$

IV.2 General solution of a second-order linear differential equation with second member

For second-order linear differential equations with constant coefficients and second member, the following theorem gives the form of the general solution.

Theorem 4.
Let a, b *and* c *be real constants, with* $a \neq 0$ *and* f *a continuous function on interval* I. *The solutions of differential equation* (E) *are functions of the form :*

$$\boxed{y_g(x) = y_h(x) + y_p(x)}$$

where

- y_h *are the solutions of the equation without a second member associated with* (E): $a\,y'' + b\,y' + c\,y = 0$.
- y_p *is a particular solution of* (E).
- y_g *is called the general solution of the equation* (E).

Remark 2.
Through this theorem, we can see that to solve the differential equation (E), *there are two steps.*

- ▷ *Solving the the equation without a second member* (H) *associated with equation* (E).
- ▷ *The determination of the particular solution of equation* (E).

IV.3 Particular solution

The second step in solving the differential equation:

$$a'' + b \cdot y' + c\,y = f(x) \qquad (E).$$

is the determination of a particular solution y_p. Two cases can be distinguished.

IV.3.1 Particular solutions associated with certain forms of the second member $f(x)$

See Course supplement 2.

Example 6.

1. *Let the following equation:*

$$y'' - y = 1. \qquad (1)$$

$y_p(x) = -1$ *being an obvious particular solution (which can also be obtained from Table* 2 *in the case where the second member* $f(x)$ *is a constant). Using the solutions of the equation without a second member associated with* (1) *determined in equation* 1 *from example* 1*, we obtain the general solution of* (1) *:*

$$\boxed{y_g(x) = y_h(x) + y_p(x) = C_1 e^{-x} + C_2 e^{x} - 1, \quad C_1,\ C_2 \in \mathbb{R}.}$$

2. *Consider the following differential equation :* $y'' + y' + y = x. \qquad (2).$

The second member of this differential equation is a polynomial of degree 1 *with* $c \neq 0$*, so according to Table* 2*, a particular solution* y_p *is a polynomial of degree* 1*. We have :*

$$y_p(x) = ax + b \Longrightarrow y_p'(x) = a, \quad y_p''(x) = 0.$$

By injecting $y_p(x)$*,* $y_p'(x)$ *et* $y_p''(x)$ *in equation* (2)*, we obtain :*

$$ax + (a + b) = x.$$

By identifying the two polynomials, we obtain :

$$\begin{cases} a = 1 \\ a + b = 0 \end{cases} \Longrightarrow \begin{cases} a = 1 \\ b = -1. \end{cases}$$

Thus, the particular solution of the equation (2) *is:* $y_p(x) = x - 1$*. according to the second equation in example* 2*, the solutions of the equation without a second member associated with* y_h *of the homogeneous equation associated with* (2) *are given by :*

$$y_h(x) = e^{-\frac{1}{2}x}\Big(C_1 \cos(\frac{\sqrt{3}}{2}x) + C_2 \sin(\frac{\sqrt{3}}{2}x)\Big), \quad C_1,\ C_2 \in \mathbb{R}.$$

Thus, the general solution of equation (2) *is given by :*

$$\boxed{y_g(x) = y_h(x) + y_p(x) = e^{-\frac{1}{2}x}\Big(C_1 \cos(\frac{\sqrt{3}}{2}x) + C_2 + \sin(\frac{\sqrt{3}}{2}x)\Big) + x - 1 \quad C_1,\ C_2 \in \mathbb{R}..}$$